

GANGAM PURNA CHAND REDDY

Embedded Systems & IoT | Software Development

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SUMMARY

Final-year Electronics & Communication Engineering student focused on embedded systems and IoT, with hands-on experience building FPGA, sensor, and microcontroller-based hardware projects. Software development experience from a full-stack school-management platform internship, structuring large-scale JSON datasets for an AI-assisted content pipeline. Comfortable working across the stack, from circuit design and firmware to backend systems.

EDUCATION

Institute of Aeronautical Engineering (IARE)

Hyderabad, India

B.Tech, Electronics & Communication Engineering (ECE) — CGPA: 6.7 | 4th Year, Expected 2027

EXPERIENCE

Software Developer Intern

May 2026 – Jun 2026

Indus Innovative Technologies Private Limited — Hyderabad, India

- Collected and structured educational content into schema-consistent JSON datasets — question banks and exam paper patterns across CBSE, ICSE, and TS SCERT board formats — for Neevpath, an AI-assisted school management platform.
- Built datasets spanning multiple subjects and grade levels (Classes 6–10), feeding the platform's automated exam paper generation system.
- Collaborated with the development team on requirement analysis, data validation, and debugging of data-parsing and rendering issues.

HARDWARE PROJECTS

FPGA-Based Image Equalizer

- FPGA-based image enhancement system built with Verilog HDL and Xilinx Vivado, processing grayscale images through hardware-based pixel and histogram processing to improve image contrast.
- Demonstrates efficient parallel image processing implemented directly in hardware logic.

Smart Battery Health Tester

- ESP32-based device that evaluates AA/AAA battery health using controlled resistive load testing rather than open-circuit voltage alone — measuring voltage sag, estimated internal resistance, and charge percentage.
- Built a non-blocking state machine, OLED UI, local Wi-Fi dashboard with REST API, and on-device calibration against a multimeter reference.

Adaptive IoT Smart Plant Irrigation System

- ESP32-based irrigation controller that predicts watering needs using OLS linear regression on soil-moisture history, rather than simple threshold switching.
- Implemented pulse-based water delivery with post-pulse verification, a full safety-focused state machine, and a self-hosted web dashboard for live monitoring and configuration.

EAGLE — Embedded Augmented Glass Lens Equipment

- Contributed to a wearable IoT smart-glass concept that syncs notifications from a mobile app via Bluetooth (Java, XML, Firebase).

SOFTWARE PROJECTS

NestHostel

- A hostel management web application handling student records and hostel operations (JavaScript).

TECHNICAL SKILLS

Embedded & Hardware: ESP32, Arduino, Verilog HDL, Xilinx Vivado, Sensors, Circuit Design, IoT

Programming: C/C++, Python, JavaScript, Java

Software Development: Django, Flutter, REST APIs, JSON, Git/GitHub

Tools: VS Code, PlatformIO, LittleFS